

Presentation

Building a Movie Recommendation System

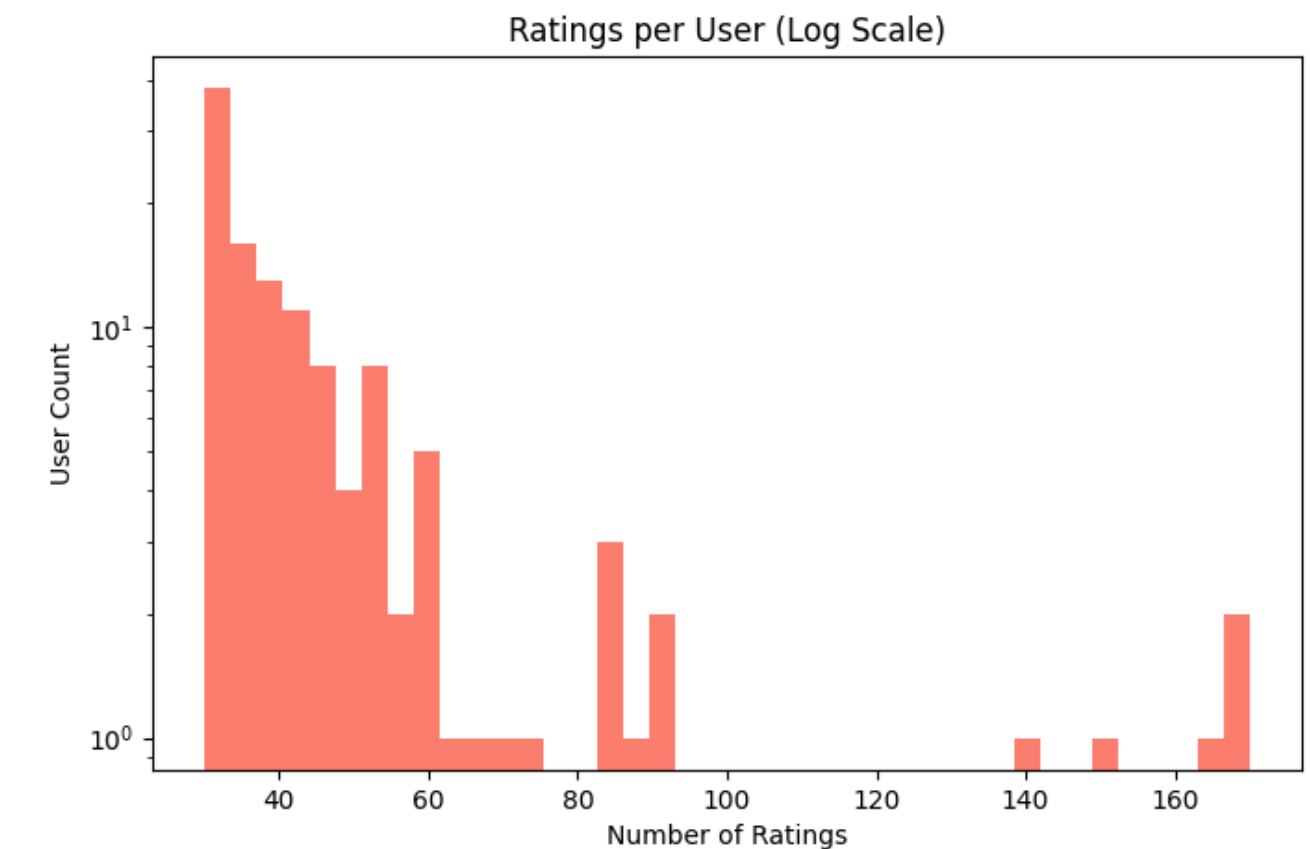
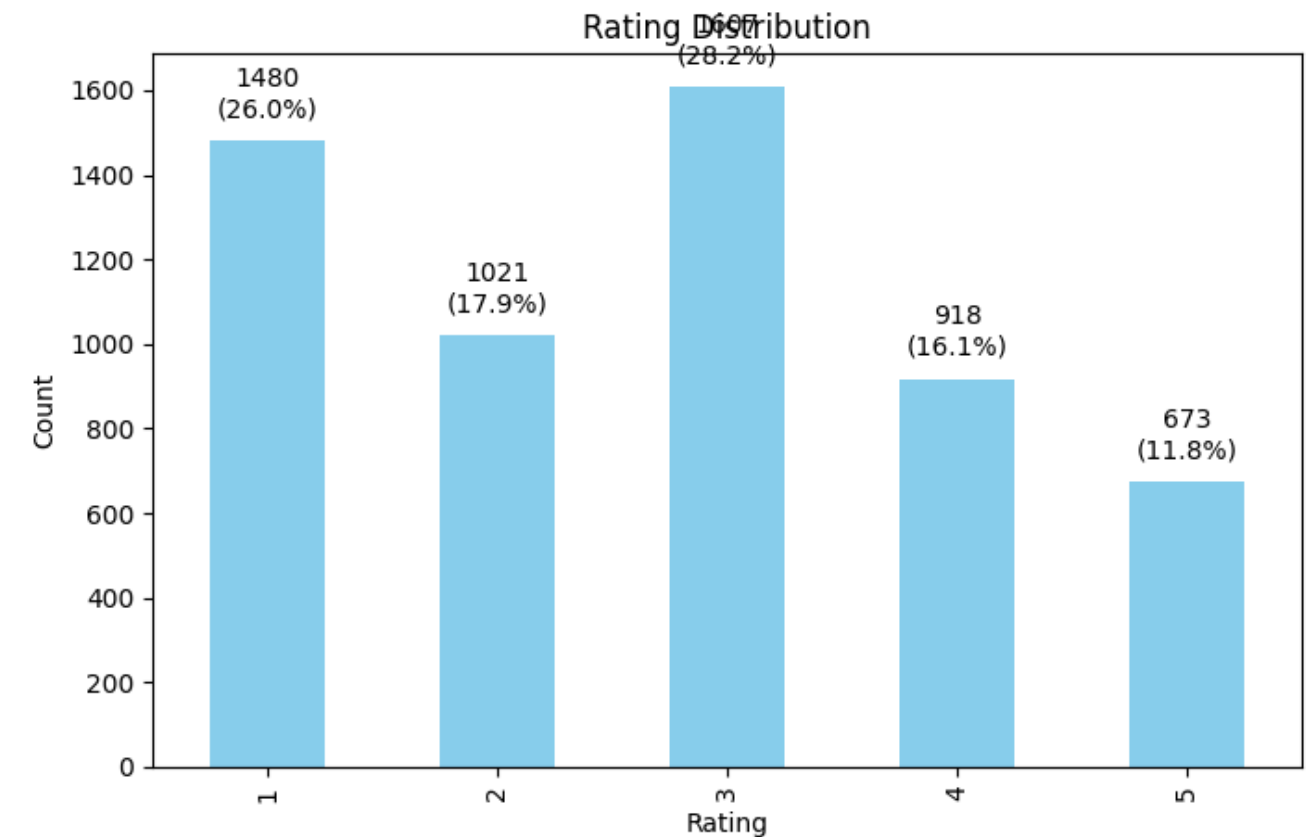
Optimizing latent factor models for personalized content delivery

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- **The Sparse Data Challenge:** Platforms like Netflix have vast movie catalogs, but users only rate a tiny fraction of items.
- **Predictive Accuracy:** We need to accurately estimate user ratings (1–5 scale) for unseen movies to provide personalized top-K recommendations.
- **Information Overload:** Guiding users to relevant content prevents decision fatigue and increases platform engagement.
- **Scale & Efficiency:** The solution must balance high-quality prediction metrics with memory-efficient training constraints.

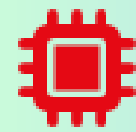


Our Approach



1. Data Filtering

Aggressively parsed the raw dataset to filter out inactive users (< 30 ratings) and unpopular movies (< 50 ratings) to reduce noise and optimize memory.



2. SVD Modeling

Trained a Singular Value Decomposition (SVD) algorithm via scikit-surprise, isolating 20 latent factors to map users and items into a shared feature space.



3. Evaluation

Implemented an 80/20 train-test split. Evaluated the matrix factorization using strict metrics including RMSE, MAE, and an efficient MAP@10 implementation.

Key Exploratory Insights

72.8%

Matrix Sparsity

- **The Long Tail:** Even after filtering active accounts, 74.2% of users have fewer than 50 ratings.
- **Rating Distribution:** The mean movie rating is 2.70 out of 5. Users are highly critical, with only 27.9% of all ratings being 4 or higher.
- **Significance:** The high sparsity (72.86%) proves that standard correlation methods fail. Latent factor modeling is strictly necessary to infer missing connections.

Experimental Results

Metric	Achieved Value	Definition
RMSE	0.9673	Root Mean Squared Error: Average rating prediction deviation.
MAE	0.7545	Mean Absolute Error: Average absolute prediction error.
MAP@10	0.6171	Mean Average Precision: Evaluates ranking quality of top-10 lists.
Catalog Coverage	29.71%	Percentage of total catalog effectively recommended across users.

Success Case:

Actual Top Rated Movies

The user displayed a strong affinity for specific action, comedy, and intense drama titles, rating them 5/5.

- Smokey and the Bandit Part 3
- Something's Gotta Give
- Richard Pryor: Live on the Sunset Strip
- Spartan
- Justice League

SVD Top-10 Predictions

The latent factors successfully identified overlapping genres, recommending high-quality, relevant matches.

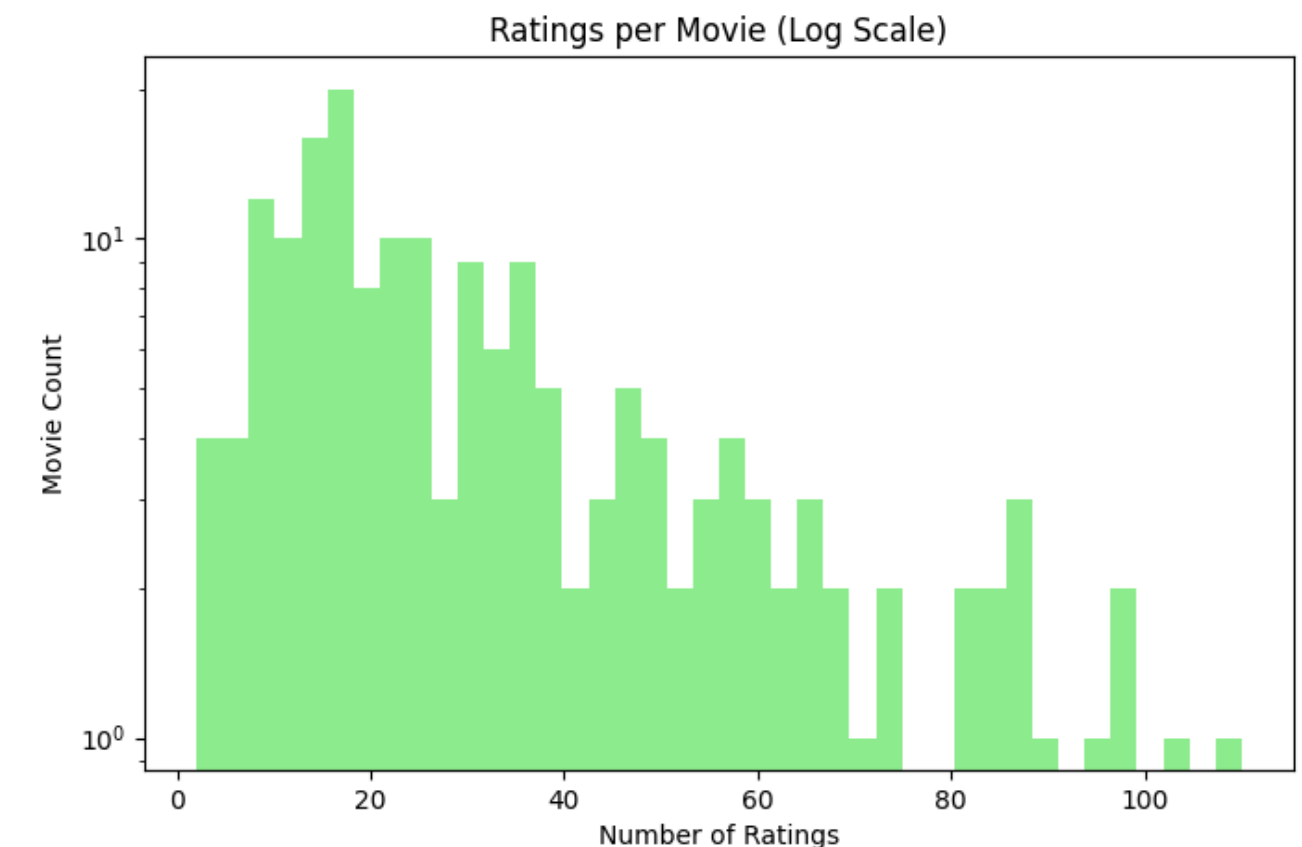
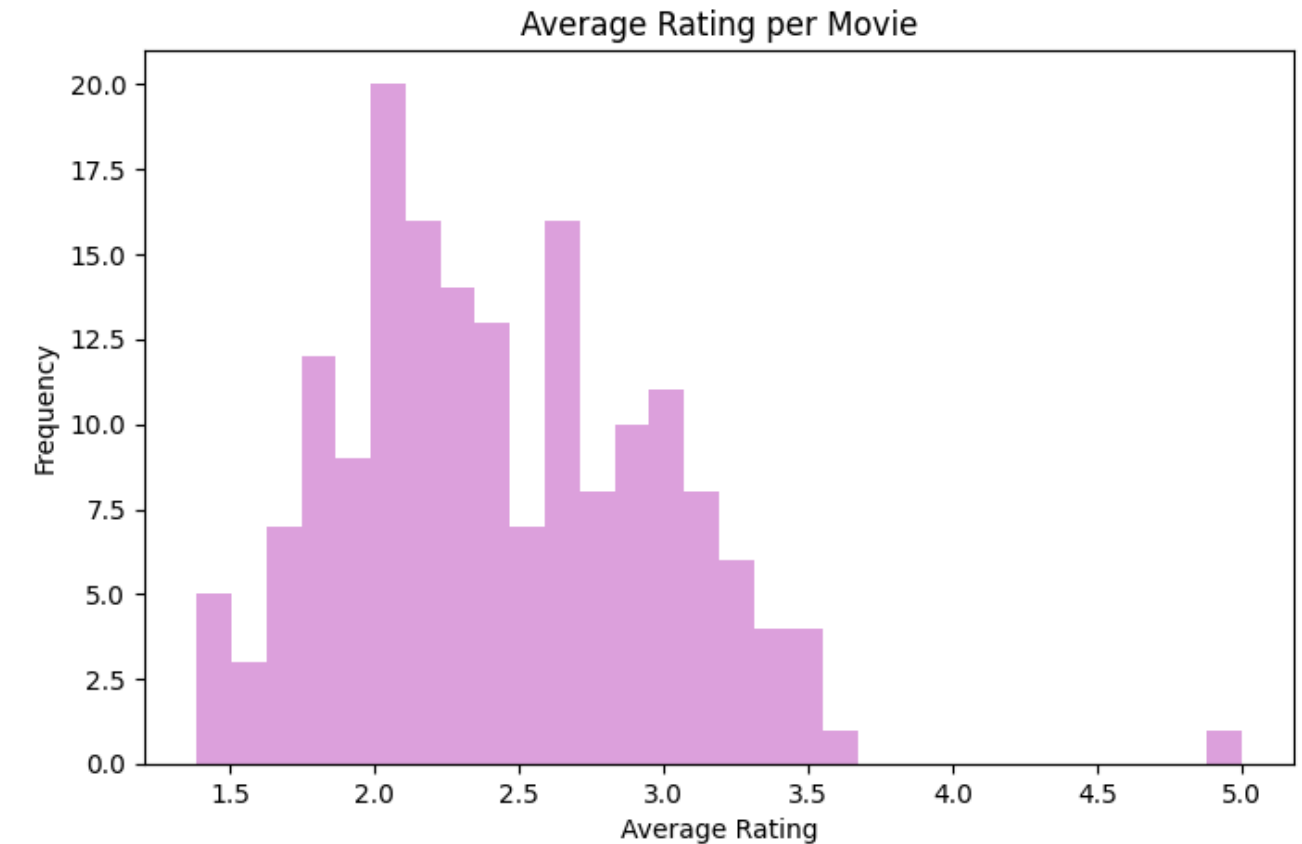
- Reservoir Dogs (Pred: 3.95)
- The Passion of the Christ (Pred: 3.82)
- Rudolph the Red-Nosed Reindeer (Pred: 3.78)
- Funny Face (Pred: 3.71)
- The Killing (Pred: 3.52)

Failure Cases & Limitations

In our failure analysis, this user exclusively liked movies with broken metadata or highly niche labels (e.g., "Unknown" entries).

The Outcome: The algorithm defaulted to generating universally popular recommendations (e.g., "Reservoir Dogs", "Elfen Lied") rather than personalized hits.

Key Insight: SVD struggles with the "cold-start" problem and ambiguous metadata. We need rich item content features (hybrid filtering) to solve this limitation.



THANK YOU!

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